



ग्रेटर नोएडा इंस्टीट्यूट ऑफ टेक्नोलॉजी (इंजीनियरिंग इंस्टीट्यूट)
GREATER NOIDA INSTITUTE OF TECHNOLOGY (Engg. Institute)



Synopsis

on

INTERACTIVE DASHBOARDS

AND VISUAL ANALYTICS : ENHANCING DECISION MAKING IN BUSINESS

INTELLIGENCE

SUBMITTED IN PARTIAL FULLFILLMENT FOR AWARD OF DEGREE OF MASTER OF
COMPUTER APPLICATIONS

IN

DEPARTMENT OF COMPUTER APPLICATIONS

BY

Aniket Kushwaha (2401320140028)

UNDER THE GUIDANCE

OF

Mr. Harendra Singh (Dy HOD - DCA)

Dr. Abdul Alim (Asst. Prof -DCA)



DEPARTMENT OF COMPUTER APPLICATIONS

Greater Noida Institute of Technology (Engg. Institute), Greater Noida

Dr. A.P.J. Abdul Kalam Technical University, Lucknow

DECLARATION

We hereby declare that our Project Synopsis Report titled “**Interactive Dashboards and Visual Analytics: Enhancing Decision Making in Business Intelligence** ” is our original contribution to the project work carried out under the guidance of **Mr. Harendra Singh** in partial fulfillment of the credit requirements for the degree of **Master of Computer Applications**. This project report has not been copied or duplicated from any other paper, journal, document, or book.

This is an authentic piece of work, and in case of any queries regarding the same, we shall be held responsible for providing clarification. Furthermore, we declare that all information presented in this document has been obtained and documented in accordance with academic rules and ethical standards.

(2401320140028, Aniket Kushwaha)

Date: 03/12/2025

Place: Greater Noida

ACKNOWLEDGEMENT

We sincerely extend our heartfelt gratitude to **Mr. Harendra Singh** of the **Department of Computer Applications** and **Dr. Abdul Alim** for their invaluable contributions to the successful completion of our project titled **“Interactive Dashboards and Visual Analytics: Enhancing Decision Making in Business Intelligence.”**

Your valuable guidance, insightful suggestions, and continuous support were immensely helpful throughout the project, and for this, we are truly grateful. We would also like to express our sincere appreciation to our parents and friends for their unwavering support and encouragement, which proved to be invaluable during various phases of the project.

(2401320140028, Aniket Kushwaha)

Abstract

In today's data-driven business environment, organizations generate volumes of information but often struggle to convert it into meaningful insights. This project, *Interactive Dashboards and Visual Analytics: Enhancing Decision-Making in Business Intelligence*, aims to bridge that gap by transforming raw business data into intuitive, interactive, and visually rich dashboards. Using a structured dataset containing sales, customer behavior, product performance, and regional metrics, the project leverages modern Business Intelligence (BI) tools to simplify complex data patterns and provide real-time, actionable insights.

This project explores the use of Data Visualisation techniques, including dashboards, heatmaps, forecasting, trend and anomaly detection, to identify trends and future scope efficiently. The proposed approach enhances prediction accuracy compared to conventional methods and provides a scalable solution for real-time knowledge of sales, customer engagement and future scopes.

Keywords: Data visualisation; forecasting; bi dashboards; heatmaps; sales insights; real time reporting; data driven insights.

Index

S. No.	Topic	
a	Declaration	
b	Acknowledgement	
c	Abstract	
1	Introduction	
1.1	Implementation	
1.2	Data Preprocessing	
2	Methods	
3	Algorithm	
4	Model	
5	Conclusion	
6	References	

Introduction

1.1 Background

In the era of data-driven decision-making, businesses increasingly rely on Business Intelligence (BI) tools to transform raw data into actionable insights. Among these tools, Interactive dashboards and visual analytics have emerged as critical components for enhancing organizational decision-making. This project explores the role of interactive dashboards and visual analytics in improving the efficiency, accuracy, and speed of business decisions.

Visual Analytics is done in Descriptive Analysis, which is one of the four types of Data Analytics Categories. This involves the use of visuals such as charts, graphs and figures to represent the data insights. It is basically a process to represent the story telling for our dataset.

An **interactive dashboard** is a visual interface that displays key performance indicators (KPIs), charts, and metric in a single screen. Unlike static reports, interactive dashboards allow users to **filter, drill down, hover for details**, and **change parameters** in real time. This helps users explore data based on their needs.

1.2 Problem Statement

There is a clear need for an **interactive, user-friendly, and real-time visual analytics dashboard** that transforms raw data into intuitive, actionable insights. This project aims to solve the problem by designing and implementing a BI-driven dashboard that enhances decision-making, improves operational efficiency, and empowers businesses to make strategic, informed, and faster decisions.

1.3 Objectives

This research aims to:

- To develop an interactive business intelligence dashboard

- To transform raw datasets into meaningful insights by using data processing and visualization techniques to convert unstructured information into actionable metrics and KPIs.
- To enhance data-driven decision-making enable managers and business stakeholders to make faster and more accurate decisions using real-time visual analytics.
- To improve trend and pattern identification by using visualization tools to help users easily spot trends, anomalies, correlations, and performance gaps.

1.4 Scope of Study

The scope of this study focuses on the design, development, and evaluation of interactive dashboards to improve the decision-making process in business environments. The key areas covered within this project include:

- Collecting, cleaning, and preparing business datasets.
- Creating visual dashboards using tools like Power BI/Tableau.
- Applying visual analytics (charts, KPIs, trend lines, filters, drill-downs).
- Improving decision-making through clear data insights and comparisons.

Implementation

2.1 Data Collection and Preprocessing

The implementation of **Interactive Dashboards and Visual Analytics Enhancing Decision making in Business Intelligence** begins with data collection from reliable sources such as **Kaggle , GFG and publicly available datasets**. The dataset typically includes the following attributes:

- **Personal Details:** user id, user name
- **Activity Features:** Activity type (review, about product), product link, image link, product link etc.
- **Expenditure:** actual price, discounted price, rating count.

2.1. Data Preprocessing

Before training the model, raw data undergoes several preprocessing steps:

- **Handling Missing Values:** Filling missing values using mean/mode imputation or dropping incomplete records.
- **Feature Encoding:** Converting categorical variables (e.g., activity type) into numerical representations using **One-Hot Encoding** or **Label Encoding**.
- **Feature Scaling:** Normalizing or standardizing numerical features to ensure uniformity, improving model convergence.

2.2. Define Relationships

If your dataset has multiple tables:

- Sales → Customers
- Products → Categories
- Orders → Payments

2.3 Dashboard Designing

Create Visualizations

Use

:

- Line charts (trends)
 - Bar charts (comparisons)
 - Pie charts (distribution)
 - Maps (geographical insights)
 - KPI cards
 - Slicers/filters
-

Methods

1. Data Acquisition

The dataset used in this project is **amazon.csv**, which contains transactional and operational data from an e-commerce environment.

This dataset includes fields such as:

- Product Information
- Category
- Selling Price / MRP / Discount
- Ratings / Reviews
- Seller Details
- Shipping details
- Profit-related attributes

This dataset represents real-world retail operations and helps in building KPIs like sales performance, pricing analytics, product rankings, and vendor performance

2. Data Preprocessing

Preprocessing was done to improve data quality and prepare for visualization.

2.1 Handling Missing Values

- Identify missing/NULL values.
- For numerical missing values → use **mean/median imputation**.
- For categorical missing values → use **mode imputation**.
- If columns have >40% missing, decide to drop or transform.

2.2 Data Cleaning

- Removed duplicate rows.
- Trimmed white spaces in product names/categories.

- Standardized column names (lowercase, underscore).
- Converted currency fields to numeric formats.
- Cleaned inconsistent values (YES/No → Yes/No).

2.3 Data Type Conversion

- Dates → datetime
- Prices, discount %, ratings → numeric
- Category, seller → categorical

2.4 Outlier Detection

Using:

- **IQR** method for price, discount, and sales
- **Z-score** for ratings

Outliers that represented incorrect data were removed; genuine business outliers (e.g., high-value products) were retained.

3. Exploratory Data Analysis (EDA)

The EDA step extracts preliminary insights:

3.1 Univariate Analysis

- Distribution of ratings
- Price distribution
- Discount frequency
- Count of products per category

3.2 Bivariate Analysis

- Price vs Discount
- Ratings vs Sales

- Category vs Avg Selling Price

- Category vs Profit

3.3 Multivariate Analysis

- Correlation heatmap
- Scatter plot matrix (price, rating, discount, sales)

This helps identify the most profitable categories and the factors influencing customer choices.

4. KPI Identification

For the dashboard, key performance indicators were selected:

4.1 Sales KPIs

- Total revenue
- Average price
- Category-wise sales

4.2 Product Performance KPIs

- Best-selling products
- Products with highest ratings
- Products with maximum discount

4.3 Profitability KPIs

- Margin per category
- Highest vs lowest profit products

4.4 Customer Insights

- Distribution of ratings
- Review sentiment (if text is provided)

5. Dashboard Design & Development

The project implements a multi-layer interactive dashboard using **Power BI**

The design follows BI best practices:

5.1 Layout Structure

1. Home Overview Dashboard

- KPIs
- Total revenue
- Average discount
- Category distribution

2. Product Analytics Dashboard

- Price vs Rating chart
- Category-wise comparison
- Discount distribution

3. Seller Performance Dashboard

- Top sellers

- Revenue by seller
- Seller-wise ratings

4. Profitability Dashboard

- Profit heatmaps
- High-margin vs low-margin products

5.2 Interactivity Features

- Dynamic Filters: category, seller, price range
- Drill-down: Category → Product → Seller
- Slicers for time, rating, discount
- Hover tooltips
- Bookmarks for story-based navigation

5.3 Visualizations Used

- **Bar charts** (top categories)
- **Line charts** (pricing trend)
- **Pie/Donut charts** (product composition)
- **Heatmaps** (correlation, profit distribution)
- **Scatter plots** (rating vs price)

- **Tables & cards** for KPIs
-

6. Business Intelligence Analytics

Advanced BI techniques were applied:

6.1 Trend Analysis

Identifies

- Categories with growing or declining demand
- Price fluctuations
- Patterns in customer ratings

6.2 Comparative Analytics

- Best vs worst sellers
- High vs low-performing categories
- Discount impact on ratings and sales

6.3 Forecasting (Optional)

Using models such as:

- Linear Regression

Forecast future sales or prices.

7. Deployment

Power BI

- Published to Power BI Cloud
- Scheduled data refresh

Jupyter

- Plots
- Heatmaps
- Scatter plot

7. Evaluation

Dashboard evaluation metrics

- Responsiveness
- KPI accuracy
- User interactivity
- Clarity of insight load time

User feedback was taken from sample business analysts to confirm usability and business relevance.

Model

4.1 Machine Learning Models Considered

- Linear Models: Provide a baseline but may not capture complex relationships.
- Decision Trees and Random Forests: Suitable for handling feature interactions and non-linearity.
- Dash boards and KPI in Powerbi.

The Random Forest Regressor was found to be the most effective due to its high accuracy, robustness, and ability to handle both linear and non-linear patterns in the dataset.

4.2 Model Optimization Strategies

- Hyperparameter Tuning: Grid Search and Bayesian Optimization to find the best model configuration.
- Feature Engineering: Selecting the most relevant features to improve predictive accuracy.
- Cross-Validation: k-fold validation to ensure the model generalizes well to new data.

4.3 Deployment and Real-Time Integration

The final model is deployed as a dashboard on

- Powerbi
- Jupyter Notebook

Conclusion

This research presents a machine learning-based approach for calorie burnt prediction, which provides an accurate and efficient estimation of energy expenditure. The study evaluated multiple machine learning models and identified the Random Forest Regressor as the most effective technique due to its ability to capture complex relationships between physiological parameters and activity-based data.

The proposed model enhances fitness tracking systems by providing personalized calorie expenditure insights. The deployment of the model in wearable devices and mobile applications allows users to monitor their daily activities and optimize their fitness goals.

Future Work

Future enhancements to this research include:

In the future, this dashboard can be improved by adding real-time data updates, predictive analytics, and AI-based recommendations. Natural language queries and voice-based insights can make the system more user-friendly. Mobile optimization and cloud deployment will enhance accessibility and scalability. Stronger security, automated report generation, and integration with more data sources can further improve the overall efficiency and intelligence of the BI system.

This research contributes to the field of health informatics and fitness technology by providing a data-driven solution for accurate Business intelligence.

References

Books & Academic Literature

1. **Shneiderman, B. (1996).** The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations. IEEE Symposium on Visual Languages.
2. **Munzner, T. (2014).** Visualization Analysis and Design . CRC Press.
3. **Few, S. (2006).** Information Dashboard Design: The Effective Visual Communication of Data. O'Reilly Media.

Research Papers & Journals

4. **Heer, J., Bostock, M., & Ogievetsky, V. (2010).** A Tour through the Visualization Zoo. ACM Queue.

Technical Documentation (Tools Used in the Project)

5. **Microsoft Power BI Documentation. (2024).** Power BI Overview.
Available at: <https://learn.microsoft.com/power-bi/>

6.Jupyter Notebook

Datasets (If Needed)

- 7..Kaggle Datasets. (2024). Public Datasets for Analysis. Available at:
<https://www.kaggle.com/datasets>

Industry & Practical Sources

- 8.Gartner. (2023). Modern BI and Analytics Platforms: Market Guide. Gartner Research.
- 9.McKinney, W. (2017). Python for Data Analysis (2nd ed.). O'Reilly Media